

International AS and A-level

Physics (9630)

PH01 Mechanics, materials and atoms

Report on the Examination

January 2025

REPORT ON THE EXAMINATION: INTERNATIONAL AS PHYSICS (9630) PH01 MECHANICS, MATERIALS AND ATOMS – JANUARY 2025

The paper was of a similar demand to those of previous series. The mean mark decreased slightly from the previous series but was more in line with the performance in January 2024.

As in previous series the calculations were well answered by most students. For some of the more demanding multi-step calculations, many students were able to perform some of the initial steps but often made an error, which resulted in them not arriving at the correct final answer. Students were awarded marks for correct initial steps even if the calculation was incomplete or the final answer was incorrect.

Generally, students found questions requiring an explanation to be the most demanding. Where students were not awarded marks for their explanations it was usually due to insufficient detail in their written answers.

Question 10 in this paper required students to process some data by plotting a graph and determining the gradient and the intercept using the equation of the line of best fit. Some students used the function on their calculator to determine the equation for a line of best fit, without drawing the line, and used that to determine the gradient and intercept. Unfortunately, just inputting the values from the table into a calculator prevents the awarding of some of the marks, as the skills that are being assessed are not demonstrated by the students.

QUESTION 01

Approximately two-thirds of students were able to give one conclusion from the Rutherford scattering experiment. Most of those students expressed the idea that because the volume of the nucleus was much smaller than the volume of atom, the atom was mostly empty space. Stating that the nucleus was very small was not enough for a mark; the size of the nucleus needed to be compared to the size of the atom. 25% of students were awarded 2 marks on this question.

QUESTION 02

This question discriminated well and gave a reasonable spread of marks. 65% of students were awarded 1 mark for attempting to determine a reasonable area under the graph. The great majority of those students also scored the second mark for correctly determining both areas, but then added the two values rather than subtracting. Approximately 40% of students were awarded full marks. Some students used the correct method and subtracted the two values but were not awarded full marks due to misreading the scale.

QUESTION 03

This question required students to explain any changes to the forces on an aircraft as it accelerates along a runway and takes off. Some students struggled to answer as they did not answer in terms of both the acceleration and the take-off.

Explaining what happens to the lift was the most awarded mark, as the mark was awarded even when it was not clear whether the student was referring to both situations. This resulted in nearly 80% of students being awarded at least one mark. Many students wrote contradictory answers to explain the change in the contact force; many thought that it would increase as the aircraft accelerated. When explaining the change in drag most students failed to add the detail that drag increases with speed. Only 15% of students were awarded all three marks.

QUESTION 04

Question 04.1 was well answered with over 80% of students gaining at least one mark. Most students were able to determine A and Z for nucleus X . Nearly 70% of the students were able to determine the specific charge of one of the nuclei, but approximately 15% of students did not go on to determine the difference.

Question 04.2 discriminated well. Approximately 50% of the students were able to gain a mark for either stating that the two products moved in opposite directions or that momentum was conserved. Nearly 40% of students made both these points. Only 10% of students gained full marks by explaining how the relative masses of the two products affected the magnitude of the velocities. Very few students wrote answers in terms of energy.

QUESTION 05

The two calculations in question 05 were very accessible. Nearly 60% of students gained full marks on 05.1 and more than 70% of students gained full marks on 05.2.

The most common error in question 05.1 was an error in trigonometry, where students used the incorrect function.

In question 05.2, the vast majority of students recognised the need to round down the number of bags, with only a small proportion rounding up.

QUESTION 06

As expected, question 06.1 produced a good spread of marks, but only approximately 15% of students were awarded 4 or 5 marks. The most common reason why students were not awarded 4 or 5 marks was due to their answer not detailing the measurement needed for accuracy. Very few students stated the actual number of distances that would be used, or suggested what these distances should be. This is the level of detail that is expected of students at AS/A-level. Over 50% of students gained 1 or 2 marks, so had a basic understanding of the procedure, but were unable to write a coherent description.

Question 06.2 discriminated well. Students who failed to score any marks did not attempt to determine the half-life from the graph and tried to read the time off the graph for a count rate that was $1/16$ of the original value. No credit was given for doing this as the value is not on the graph. Very few students determined more than one value for the half-life to find an average, so only approximately 5% of students scored full marks.

QUESTION 07

Question 07.1 was quite an accessible calculation with nearly 50% of students gaining full marks. Nearly 30% of students failed to score any marks, and many of those tried to take a conservation of momentum approach to the calculation. Most students who performed a correct initial step were able to go on to arrive at the correct final answer.

Question 07.2 was reasonably well answered, but approximately 30% of the students who correctly identified that B had the greater magnitude of acceleration, failed to write a detailed explanation as to why that was the case. Many students just stated the difference in mass.

QUESTION 08

Question 08.1 was more demanding than expected. Many students wrote simple statements that the height or acceleration was the same for both. Only a little over 20% of students wrote a complete explanation that included that the initial vertical component of both velocities was the same.

Question 08.2 was also more demanding than expected with just under 50% of students being awarded the mark. Lots of students gave 1 significant figure answers and were not awarded the mark.

Question 08.3 discriminated very well and provided a good spread of marks. However, the vast majority of students did not draw a scale drawing. Most did draw a vector triangle, but there was no scaling and often a ruler was not used. For marks to be awarded the scale used had to be stated, and the diagram had to be drawn accurately with a ruler. Lots of students did score marks, as up to two marks could be awarded for a correct algebraic approach.

Questions 08.4 and 08.5 performed as expected, with very few students scoring full marks. The first mark in both parts was reasonably accessible so approximately 40% of students were able to gain one mark on both questions.

QUESTION 09

Question 09.1 was one of the most accessible questions on the paper with over 80% of students correctly completing the equation.

Question 09.2 was one of the most discriminating questions on the paper and produced a good spread of marks. 65% of students gained at least one mark for reading the value off the graph. However, less than half of these went on to arrive at the correct final answer. Some students failed to convert to joules, and some struggled to rearrange the kinetic energy equation. These are skills that are regularly assessed.

Just over 5% of students scored full marks on question 09.3. Nearly 50% did score 1 mark, but many of these did not state that the same amount of energy is released in each decay.

Question 09.4 was more demanding than expected. There was some judgement required to read off the maximum value, but most students who misread this value did not realise that the minimum energy was zero.

QUESTION 10

Question 10.1 was well answered with most students correctly equating the moments.

Question 10.2 was reasonably well answered; the points were not difficult to plot, as they were on major y-axis grid lines. Where points were not plotted accurately, the mark was not awarded.

In Question 10.3, the mark was only awarded if the scatter was considered, and the line was a continuous straight line through the origin.

Question 10.4 was correctly answered by approximately 40% of students. Where the intercept was not read, very few students were able to successfully calculate it using the equation of the line and points from the line.

In question 10.5, similarly to question 10.4, students who attempted to use the equation of the line and points from the line often made a mistake and did not score full marks. This method often included using a point from the table that was not on their line of best fit. Most students who answered well drew a gradient triangle, although some lost a mark due to the triangle being too small.

QUESTION 11

Question 11.1 was one of the most discriminating questions on the paper. Of all the calculations on the paper, it produced the best spread of marks. Just under 40% of students completed all the steps correctly and arrived at the correct final answer. 40% of students completed some steps correctly and were evenly spread about 1, 2 or 3 marks. The common errors were incorrectly converting from MPa, using the incorrect trigonometric ratio, or incorrectly rearranging the equation for the cross-sectional area.

Question 11.2 also discriminated well and produced a good spread of marks. Lots of students demonstrated some understanding, but often lacked the detail required. They gave the correct relationship between the force and the angle, but did not explain what happened when the angle decreased.

Question 11.3 was a demanding question, with under 5% of students understanding that it would reduce the force on the climber. Most did not identify that the extension would increase.

SECTION B

The multiple-choice questions covered parts of the specification not covered in Section A.

The most accessible question was question 23 with just over 80% of students answering correctly.

Questions 12, 13, 18, 22 and 25 were also well answered with approximately 70% answering correctly.

The most challenging question was question 24 where fewer than 20% gave the correct response.